

IBM - VIT- Cybersecurity

ASSIGNMENT - 2

Web security Monitoring

Ragini Balasundaram [24BYB1115]

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Introduction

Implementation of **WEB SECURITY MONITORING SYSTEM** using **SPLUNK - cloud platform** to detect suspicious activities, track HTTP errors, and identify potential brute-force attacks.

1. Creating a dashboard with key security metrics.
2. Scheduling a daily report for HTTP errors.
3. Setting up an alert for 404 error spikes.

Objective:

- Monitor suspicious HTTP Errors (404, 500).
- Track traffic patterns (like Top IPs , error trends).
- Detect Brute- Force/ scanning attempts (excessive 404s from single IPs).

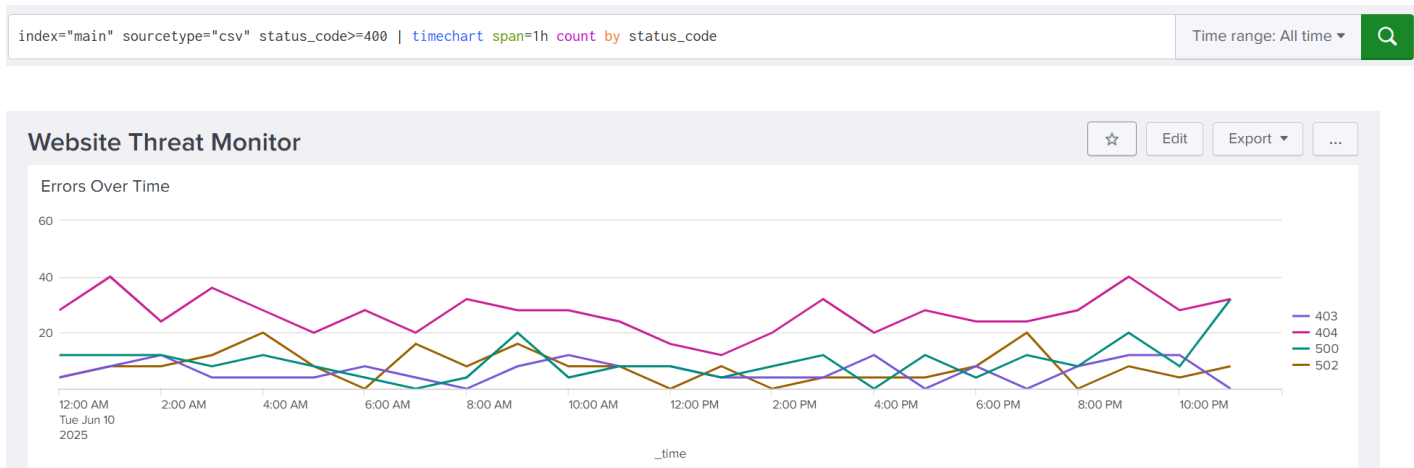
The screenshot shows the 'Add Data' wizard in the Splunk Cloud interface. The wizard is at the 'Review' step, with a progress bar indicating the sequence: Select Source, Set Source Type, Input Settings, Review, and Done. The 'Review' step is currently active. Below the progress bar, the 'Review' section displays the following configuration details:

Input Type	Uploaded File
File Name	complex_access_log.csv
Source Type	csv
Host	si-i-01adb4d0863181cbc.prd-p-dj3m9.splunkcloud.com
Index	main

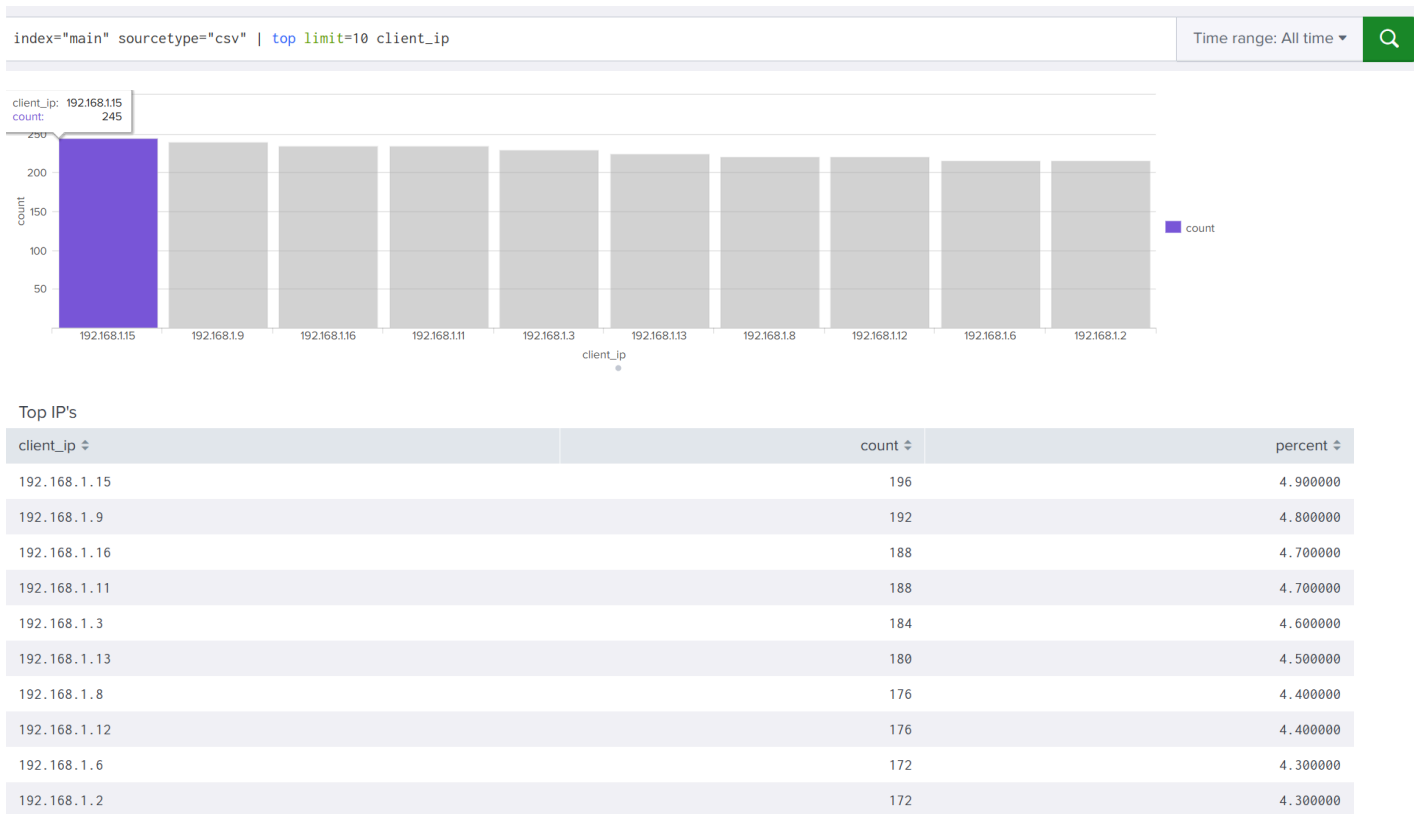
At the top of the wizard, there is a navigation bar with the Splunk Cloud logo, a search bar, and a user profile dropdown. The 'Add Data' title is centered above the progress bar. The 'Review' step is highlighted with a green dot, and the 'Submit' button is green, while the 'Back' button is grey.

DASHBOARD - WEBSITE THREAT MONITOR

Errors over time:



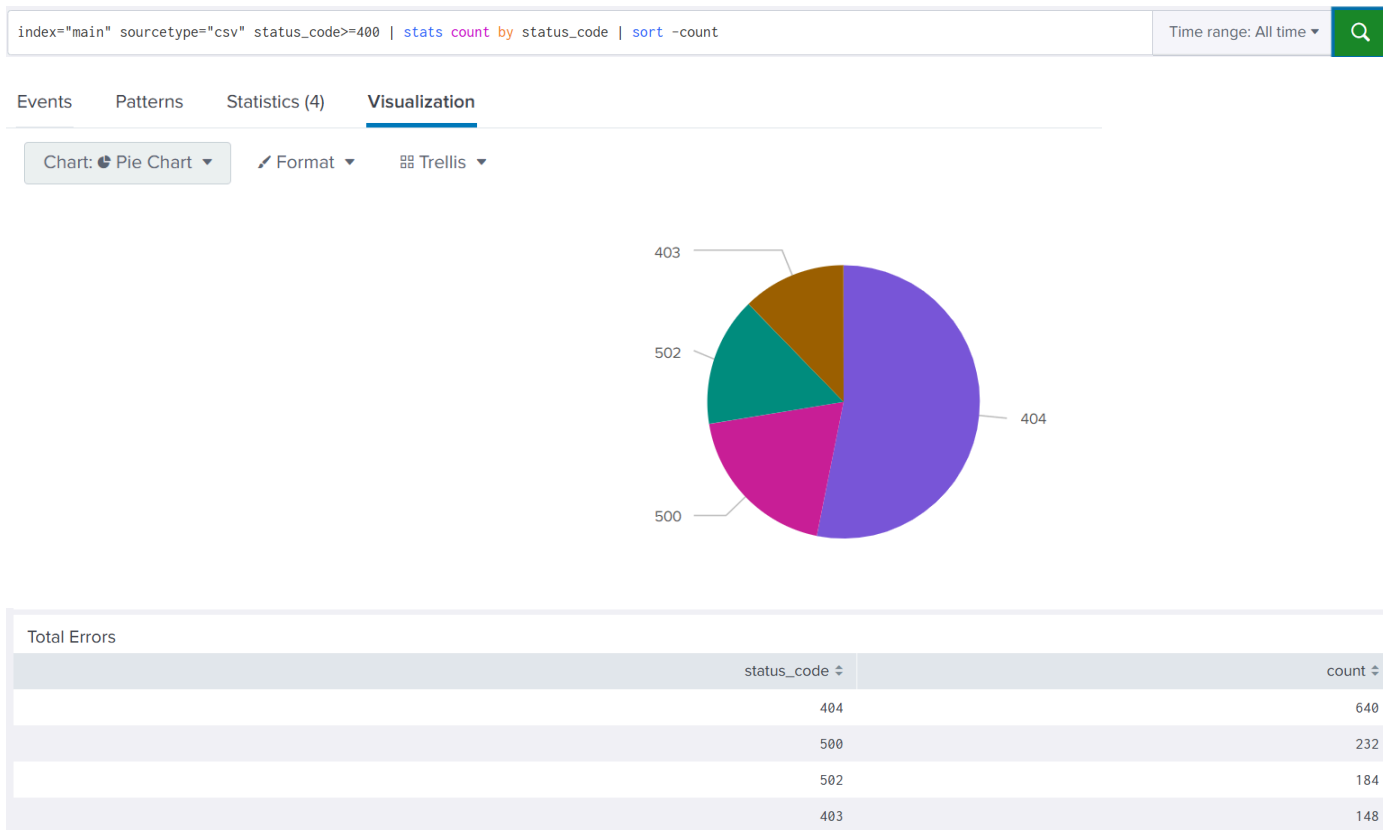
Top IPs :



High 404 IPs :

index="main" sourcetype="csv" status_code =404 stats count by clientip sort -count head 10		Time range: All time	Q
High 404 IP's			
client_ip		count	
192.168.1.6		48	
10.0.0.2		44	
192.168.1.15		36	
192.168.1.9		36	
192.168.1.17		32	
192.168.1.3		32	
192.168.1.5		32	
10.0.0.3		28	
192.168.1.11		28	
192.168.1.13		28	

Total errors :



Scheduled report

index="main" sourcetype="csv" (status_code =404 OR status_code =500) | stats count by status_code

Scheduled report configuration:

Report

Daily HTTP Error Summary

Schedule Report

☒

[Learn More](#)

Schedule

Run every day ▾

At

9:00 ▾

Time Range

All time ▸

Schedule Priority ?

Default ▾

Save As Report

×

Title

Daily HTTP Error Summary

Description

Daily summary of HTTP 404 and 500 errors for security monitoring.

Content

📄 + 📄

📄

📄

Time Range Picker

Yes

No

Cancel

Save

Daily HTTP Error Summary

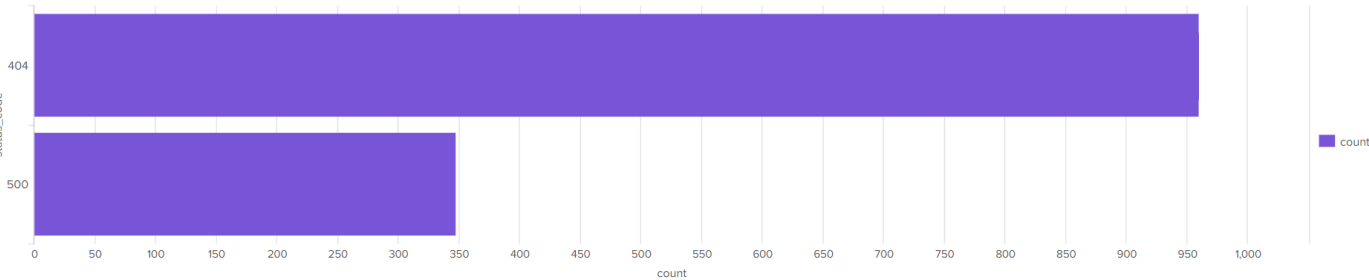
Daily summary of HTTP 404 and 500 errors for security monitoring.

All time ▾

✓ 1,308 events (before 6/19/25 9:37:40.000 AM)

Edit ▾ More Info ▾ Add to Dashboard ▾

Job ▾ || 📄 ↺ ↻ ⌂ ⬇



2 results 20 per page ▾

status_code ▾	count ▾
404	960
500	348

Setting Alert

```
index="main" sourcetype="csv" status_code=404 | stats count as error_count | where error_count > 20
```

Alert settings:

Save As Alert



Settings

Title Spike in 404 Errors

Description Optional

Permissions

Private

Shared in App

Alert type

Scheduled

Run on Cron Schedule ▼

Time Range

All time ▶

Cron Expression

*/5 * * * *

e.g. 00 18 *** (every day at 6PM). [Learn More](#)

Expires

24

hour(s) ▼

Trigger Conditions

Trigger alert when

Number of Results ▼

is greater than ▼

20

Trigger

Once

For each result

Throttle ?

☐

Trigger Actions

+ Add Actions ▼

When triggered



✉ Send email

[Remove](#)

To ragini.balasundaram157@gmail.com

Comma separated list of email addresses.

Subject

404 Error Spike Detected - {count}

The email subject, recipients and message can include tokens that insert text based on the results of the search. [Learn More](#)

Message

Alert Name: {name}

Trigger Time: {trigger_time}

We've detected {count} 404 errors in the past 5 minutes

Include

☒ Link to Alert

☒ Link to Results

☐ Search String

☐ Inline [Table](#)

☐ Trigger Condition

☒ Attach CSV

☐ Trigger Time

☒ Attach PDF

☐ Allow Empty Attachment

Cancel

Save

Results & Analysis:

- **Top IPs** panel showed frequent visitors - potential scanners.
- **Errors over time** panel showed spikes in 404/500 errors - possible attackers.
- **High 404 IPs** identified suspicious IPs :
IP address : 192.168.1.6 with -> 48 counts .
- Alert triggered during testing.